

Lab 10 — Power Transfer in AC Circuits

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Description

The purpose of this lab is to study how average power is delivered from a sinusoidal source to a load, and how the choice of load and compensating components affects the amount of power that actually arrives. The source is modeled as an ideal AC voltage source V_{in} in series with a source resistance R_s , connected to the load through a line that is modeled as a series inductance L . Because the line reactance sits between the source and the load, the classical resistive maximum-power-transfer result ($R_L = R_s$) no longer applies, and the optimum load must account for the reactance as well.

We look at two cases. In Case 1 the load is purely resistive, $Z_L = R_L$, and we are free to choose R_L . We sweep a set of resistors across roughly three decades, measure the load voltage at each, and locate the value of R_L that maximizes the delivered power. In Case 2 the load resistance is treated as fixed — as it usually is in practice, since a lamp or a heater has whatever resistance it has — and instead a shunt capacitor C is placed directly across the load terminals. We sweep a set of capacitors and find the value that maximizes the power delivered to that fixed resistor. Case 2 is repeated for three different load resistances so that we can see whether the optimum capacitance depends on the load. Throughout, the source is set to $V_{in} = 5\text{ V}$ amplitude at $f = 5\text{ kHz}$, with $R_s = 100\ \Omega$ and $L = 33\text{ mH}$.

Procedure

1. Measured the actual values of every component used before building anything: each load resistor and the $100\ \Omega$ source resistor on the DMM, each capacitor on the LCR meter, and the inductance and DC resistance of the 33 mH inductor. These measured values, not the nominal ones, were used in all calculations.
2. Built the Case 1 circuit on the PB-503A trainer: function generator output through $R_s = 100\ \Omega$, then through the 33 mH inductor, then into the load resistor R_L returning to the generator ground. Both V_{in} (generator terminal to ground) and V_L (load node to ground) share the circuit ground, so both were measured single-ended with the scope.
3. Set the generator to a 5 kHz sine wave and adjusted the amplitude to 5 V peak (10 V peak-to-peak) as read on the oscilloscope *with the circuit connected*, since the generator's $50\ \Omega$ output impedance causes the open-circuit setting to sag under load.
4. For each load resistor in the set, recorded the resistance, the peak-to-peak load voltage $V_{L,pp}$ from the scope, and the source amplitude V_{in} at that operating point. The period was confirmed from the waveform itself rather than trusting the generator dial.
5. Rebuilt the circuit for Case 2 by adding a shunt capacitor C directly across the load terminals, in parallel with R_L , starting with $R_L = 1\text{ k}\Omega$.
6. For each capacitor in the set, recorded the capacitance and the peak-to-peak load voltage. To land near the predicted optimum, the 47 nF and 100 nF capacitors were placed in series (giving roughly 32 nF) and that combination was placed in parallel with R_L .

7. Repeated the Case 2 capacitor sweep for two additional load resistances taken from the Case 1 set, one below and one above 1 k Ω , so the three sweeps could be compared on a single plot.
8. Converted every recorded voltage to an amplitude ($V_m = V_{pp}/2$) and computed the delivered average power as $P_L = V_m^2/2R_L$.

Calculations

Case 1: resistive load only

With $Z_L = R_L$ the load voltage follows from a voltage divider across the series combination of R_s , $j\omega L$, and R_L :

$$V_L = V_{in} \frac{R_L}{(R_s + R_L) + j\omega L}$$

The average power delivered to a resistor driven by a sinusoid of amplitude V_L is $P_L = V_L^2/2R_L$, since the mean square of a sinusoid is half its squared amplitude. Substituting the divider result gives

$$P_L = \frac{V_L^2}{2R_L} = \left(\frac{V_{in}^2}{2} \right) \left(\frac{R_L}{(R_s + R_L)^2 + (\omega L)^2} \right)$$

Everything here except R_L is fixed, so setting $dP_L/dR_L = 0$:

$$\frac{d}{dR_L} P_L = \left(\frac{V_{in}^2}{2} \right) \left(\frac{-R_L^2 + R_s^2 + (\omega L)^2}{[(R_s + R_L)^2 + (\omega L)^2]^2} \right) = 0 \implies R_L = \sqrt{R_s^2 + (\omega L)^2}$$

The optimum load is therefore the magnitude of the source-side impedance, not just its resistive part. With $V_{in} = 5$ V, $R_s = 100$ Ω , $L = 33$ mH, and $f = 5$ kHz:

$$\begin{aligned} \omega &= 2\pi(5000) = 31416 \text{ rad s}^{-1} \\ \omega L &= (31416)(0.033) = 1036.7 \text{ } \Omega \\ R_{L,opt} &= \sqrt{100^2 + 1036.7^2} = 1041.5 \text{ } \Omega \\ P_{L,max} &= \frac{5^2}{2} \frac{1041.5}{(1041.5 + 100)^2 + (1036.7)^2} = 5.475 \text{ mW} \end{aligned}$$

Note that the reactance dominates: ωL is more than ten times R_s , so the optimum load is set almost entirely by the line inductance.

Case 2: fixed resistive load with a shunt capacitor

Placing C in parallel with R_L and reducing the divider gives

$$V_L = V_{in} \frac{R_L}{(R_s + R_L - \omega^2 R_L L C) + j\omega(L + R_L R_s C)}$$

so that

$$P_L = \frac{V_L^2}{2R_L} = \left(\frac{V_{in}^2}{2} \right) \left(\frac{R_L}{(R_s + R_L - \omega^2 R_L L C)^2 + [\omega(L + R_L R_s C)]^2} \right)$$

Now R_L is fixed and C is the free variable. Writing $D_1 = R_s + R_L - \omega^2 R_L L C$ and $D_2 = \omega(L + R_L R_s C)$, the condition $dP_L/dC = 0$ reduces to $D_2 R_s = D_1 \omega L$, and solving:

$$R_s(L + R_L R_s C) = L(R_s + R_L - \omega^2 R_L L C) \implies C(R_s^2 + \omega^2 L^2) = L \implies C = \frac{L}{\omega^2 L^2 + R_s^2}$$

The factor of R_L cancels completely, so the optimum shunt capacitance depends only on the source and line (ω , L , R_s) and not on the load at all. Numerically,

$$C_{opt} = \frac{0.033}{[(2\pi)(5000)]^2 (0.033)^2 + (100)^2} = 30.42 \text{ nF}$$

Evaluating the power at $R_L = 1041.5 \Omega$ with this capacitor:

$$D_1 = 100 + 1041.5 - (31416)^2(1041.5)(0.033)(30.42 \times 10^{-9}) = 109.6 \Omega$$

$$D_2 = (31416) [0.033 + (1041.5)(100)(30.42 \times 10^{-9})] = 1136.3 \Omega$$

$$P_L = \frac{5^2}{2} \frac{1041.5}{(109.6)^2 + (1136.3)^2} = 9.99 \text{ mW}$$

which is very close to twice the Case 1 optimum. The shunt capacitor partially cancels the line reactance seen from the source, which is why the power roughly doubles for the same load resistor.

One consequence worth noting: with the compensating capacitor installed the network is near resonance and the load voltage can be *larger* than the source amplitude. At $R_L = 2200 \Omega$ with $C = 32 \text{ nF}$ we measured a load amplitude of 3.49 V from a source whose delivered amplitude was only 2.0 V — a magnification of about 1.7×. The scope range was set accordingly and the generator checked for clipping.

SciLab program

```
// =====
//  ENGR 2405 - Lab 10: Power Transfer in AC Circuits
//  Theoretical and measured load power for Case 1 and Case 2.
//  Theory uses the measured delivered source amplitude Vin = 2.0 V
//  (backed out from the no-capacitor points; see report).
//  =====
clear;

Vin = 2.00;          // measured delivered amplitude, V (nominal setting was
                    // higher)
Rs  = 100;           // source resistance, ohm
L   = 33e-3;         // line inductance, H
f   = 5000;          // frequency, Hz
w   = 2*pi*f;        // rad/s
XL  = w*L;           // inductive reactance, ohm

RLopt = sqrt(Rs^2 + XL^2);
Copt  = L/(w^2*L^2 + Rs^2);
mprintf("XL = %.1f ohm    RL,opt = %.1f ohm    C,opt = %.2f nF\n", XL, RLopt
        , Copt*1e9);

// ----- CASE 1 -----
RL1 = [10 47 100 330 680 1000 2200 10000];
VL1 = [0.04 0.20 0.28 0.608 1.06 1.32 1.88 2.52]; // measured amplitude,
V
PL1m = (VL1.^2)./(2*RL1);
PL1t = (Vin^2/2)*RL1./((Rs+RL1).^2 + XL^2);
mprintf("\nCASE 1\n  RL(ohm)    VL(V)    PLmeas(mW)  PLtheo(mW)    err(%%)\n");
for k=1:length(RL1)
```

```

e=100*(PL1m(k)-PL1t(k))/PL1t(k);
mprintf("%9.0f %7.3f %10.3f %10.3f %8.1f\n",RL1(k),VL1(k),PL1m(k)*1e3,
PL1t(k)*1e3,e);
end

// ----- CASE 2 -----
function P=ploadC(C,RL,Vin,Rs,L,w)
D1=Rs+RL-w^2*RL*L*C; D2=w*(L+RL*Rs*C);
P=(Vin^2/2)*RL ./ (D1.^2+D2.^2);
endfunction

// capacitance values used (F). "series" is 47nF in series with 100nF.
Cser = 47e-9*100e-9/(47e-9+100e-9);
Clist = [0 10e-9 22e-9 Cser 47e-9 113e-9];
RLset = [1000 680 2200];
VL2 = [1.320 1.380 1.440 1.360 1.260 0.696; // RL = 1000
1.046 1.134 1.209 1.225 1.163 0.633; // RL = 680
1.740 2.273 3.154 3.490 2.537 0.723]; // RL = 2200 (reordered to
match Clist)
for i=1:size(RLset,2)
R=RLset(i);
mprintf("\nCASE 2 RL=%g ohm\n C(nF) VL(V) PLmeas(mW) PLtheo(mW)
err(%%)\n",R);
for j=1:length(Clist)
pm=VL2(i,j)^2/(2*R); pt=ploadC(Clist(j),R,Vin,Rs,L,w);
e=100*(pm-pt)/pt;
mprintf("%8.2f %7.3f %10.3f %10.3f %8.1f\n",Clist(j)*1e9,VL2(i,j),pm*1
e3,pt*1e3,e);
end
end

```

Listing 1: SciLab program: theoretical and measured load power for both cases

XL = 1036.7 ohm RL,opt = 1041.5 ohm C,opt = 30.42 nF

CASE 1

RL(ohm)	VL(V)	PLmeas(mW)	PLtheo(mW)	err(%)
10	0.040	0.080	0.018	334.8
47	0.200	0.426	0.086	396.3
100	0.280	0.392	0.179	118.5
330	0.608	0.560	0.524	6.9
680	1.060	0.826	0.808	2.3
1000	1.320	0.871	0.875	-0.5
2200	1.880	0.803	0.691	16.2
10000	2.520	0.318	0.194	63.7

CASE 2 RL=1000 ohm

C(nF)	VL(V)	PLmeas(mW)	PLtheo(mW)	err(%)
0.00	1.320	0.871	0.875	-0.5
10.00	1.380	0.952	1.149	-17.1
22.00	1.440	1.037	1.460	-29.0
31.97	1.360	0.925	1.543	-40.0
47.00	1.260	0.794	1.259	-37.0
113.00	0.696	0.242	0.233	4.1

Measurements

All theoretical values below use the *measured* delivered source amplitude $V_{in} = 2.0\text{ V}$, not the nominal setting. This amplitude was backed out from the three no-capacitor points, which independently gave 2.00 V at $R_L = 680, 1000$, and $2200\ \Omega$. The nominal maximum-power design targets ($R_{L,opt} = 1041.5\ \Omega$, $C_{opt} = 30.4\text{ nF}$) are independent of amplitude and are unchanged; only the power levels scale with V_{in}^2 .

Table 1: Measured fixed parameters

Quantity	Nominal	Measured
Source resistor R_s	100 Ω	100 Ω
Line inductor L	33 mH	33 mH
Inductor DC resistance	—	34 Ω
Source frequency f	5 kHz	5.00 kHz
Delivered amplitude V_{in}	—	2.0 V (loaded)

Case 1: resistive load only

Table 2: Case 1 — load power vs. load resistance. V_L is the measured load amplitude; theory uses $V_{in} = 2.0\text{ V}$, $R_s = 100\ \Omega$, $L = 33\text{ mH}$, $f = 5\text{ kHz}$.

R_L (Ω)	V_L (V)	P_L meas. (mW)	P_L theo. (mW)	% error
10	0.040	0.080	0.018	335
47	0.200	0.426	0.086	396
100	0.280	0.392	0.179	118
330	0.608	0.560	0.524	6.9
680	1.060	0.826	0.808	2.3
1000	1.320	0.871	0.875	−0.5
2200	1.880	0.803	0.691	16.2
10000	2.520	0.318	0.194	63.7

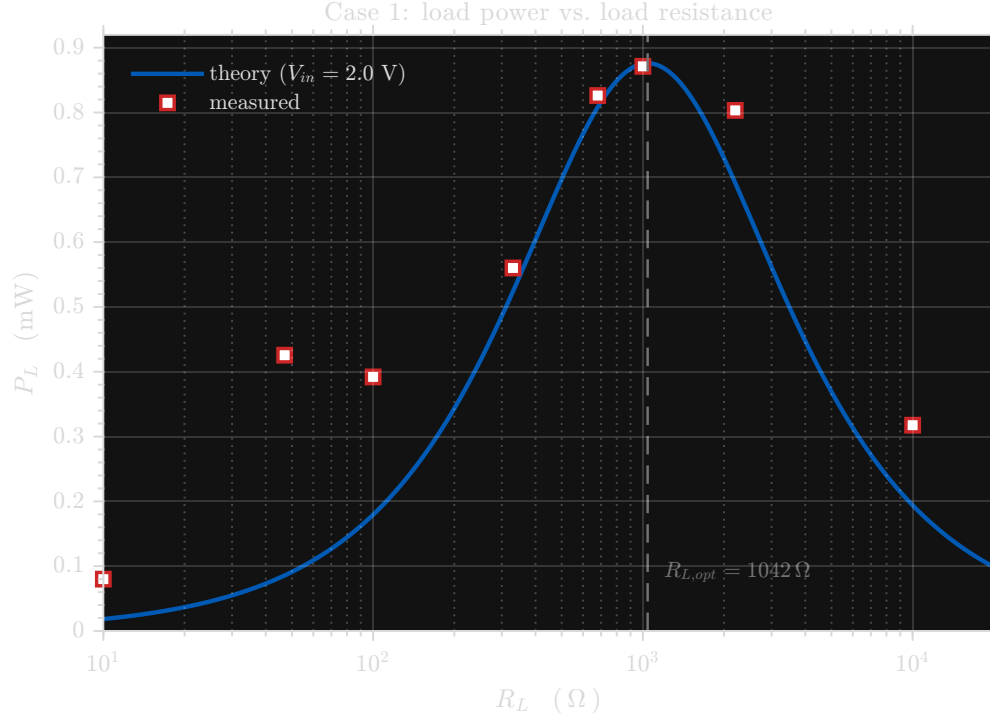


Figure 1: Case 1: measured and theoretical load power as a function of load resistance. The broad maximum sits near $R_L = 1 \text{ k}\Omega$, matching the predicted optimum of 1041.5Ω .

Case 2: fixed load with shunt capacitor

Table 3: Case 2 — measured load voltage and power vs. shunt capacitance for three load resistances. The 32 nF entry is the 47 nF and 100 nF capacitors in series.

C (nF)	V_L (V)	P_L meas. (mW)	P_L theo. (mW)
$R_L = 1000\ \Omega$			
0 (none)	1.320	0.871	0.875
10	1.380	0.952	1.149
22	1.440	1.037	1.460
32 (series)	1.360	0.925	1.543
47	1.260	0.794	1.259
113	0.696	0.242	0.233
$R_L = 680\ \Omega$			
0 (none)	1.046	0.804	0.808
10	1.134	0.946	0.950
22	1.209	1.075	1.079
32 (series)	1.225	1.103	1.109
47	1.163	0.995	0.999
113	0.633	0.295	0.296
$R_L = 2200\ \Omega$			
0 (none)	1.740	0.688	0.691
10	2.273	1.174	1.180
22	3.154	2.261	2.272
32 (series)	3.490	2.768	2.782
47	2.537	1.463	1.470
113	0.723	0.119	0.119

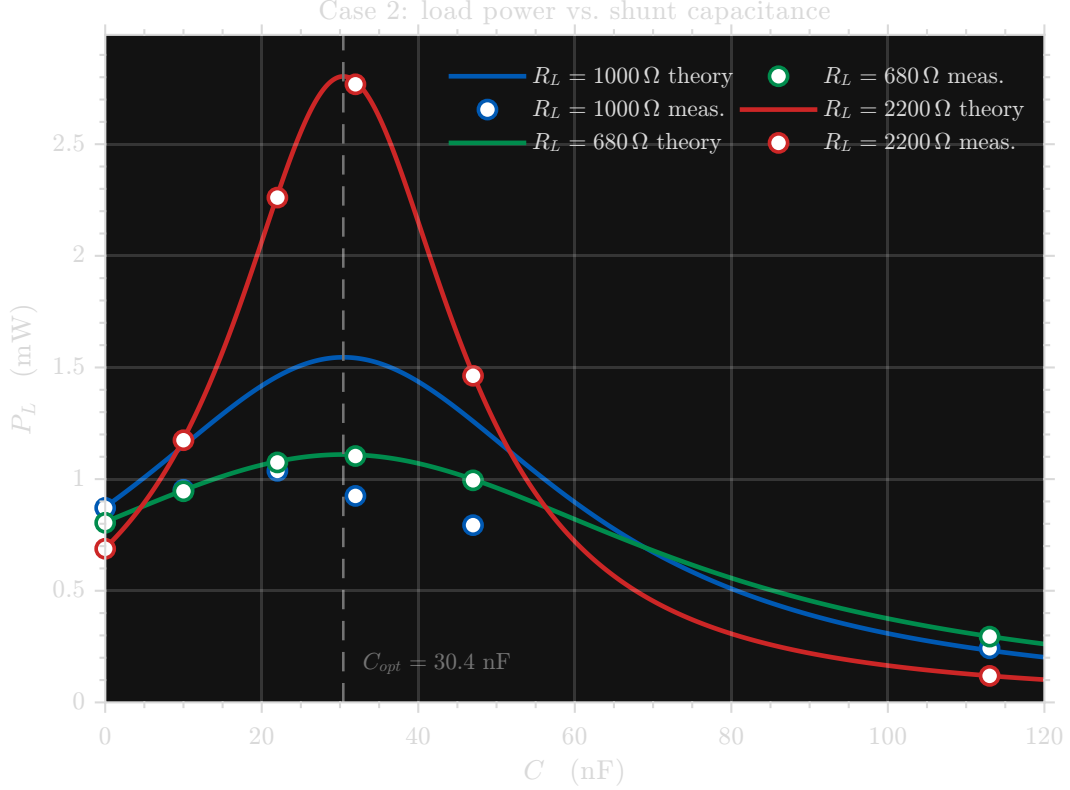


Figure 2: Case 2: measured (open markers) and theoretical (solid) load power vs. shunt capacitance for three load resistances. All three theoretical curves peak at the same $C_{opt} = 30.4\ \text{nF}$, independent of R_L .

Observations & Discussion

The delivered source amplitude was about 2.0 V, well below the nominal setting, and this is the first thing the data shows. The no-capacitor load voltages at $R_L = 680$, 1000, and 2200 Ω each imply $V_{in} = V_L |Z|/R_L = 2.0\ \text{V}$ to within a percent. The generator's output impedance sits in series with our 100 Ω resistor and the inductor's 34 Ω winding resistance, so the terminal amplitude sags once the circuit draws current, and the loaded value is the one worth comparing against. Evaluated at 2.0 V, theory predicts a peak of 0.88 mW against the 0.871 mW we measured at 1 k Ω . The optimum resistance and capacitance do not depend on amplitude, so only the power levels scale and the design conclusions hold.

In Case 1 the measured curve follows theory within about 7% from 330 Ω through 1 k Ω and peaks where it should. The maximum is very flat, with theory placing the power at 1 k Ω within a tenth of a percent of the true optimum at 1041.5 Ω , so we report peak power against prediction rather than trying to locate the peak resistance experimentally. Both ends of the sweep deviate for clear reasons. From 10 to 100 Ω the load voltage is only tens of millivolts, at the limit of the scope cursors, so the large percent errors there sit on negligible absolute powers. At 2200 and 10 k Ω the measured power runs above theory because those light loads draw less current, the source sags less, and the delivered amplitude climbs back toward its open-circuit value, which a fixed 2.0 V model cannot capture.

Case 2 shows the shunt capacitor working as intended. For every load, power rose to a maximum as capacitance increased and then fell sharply past the optimum, dropping below the no-capacitor

baseline by 113 nF. At $R_L = 1\text{ k}\Omega$ the power went from 0.87 mW to about 1.04 mW, and at $R_L = 2.2\text{ k}\Omega$ from 0.69 mW to 2.77 mW, a fourfold gain from one passive component. The capacitor draws a leading current that partially cancels the lagging current in the line inductor, so less reactive current flows through R_s and more real power reaches the load. All three theoretical curves peak at the same capacitance regardless of load, confirming that R_L cancels out of $dP_L/dC = 0$ to leave $C_{opt} = L/(\omega^2 L^2 + R_s^2)$. What changed with R_L was the height of the peak, not its position: with C_{opt} installed the best load resistance moves far above $1041\text{ }\Omega$, which is why the $2.2\text{ k}\Omega$ load beat anything achievable in Case 1.

The clearest deviation in Case 2 is at $R_L = 1\text{ k}\Omega$, where the measured peak fell at 22 nF instead of 30 nF and sat roughly a third below theory. Capacitor tolerance is the likely cause. These ceramics carry ten to twenty percent tolerances, and the 47 nF and 100 nF series pair we used to reach the optimum compounds two of them, so a true value a few nanofarads low would move the operating point down the falling side of the curve and account for both the reduced peak and its leftward shift. Capacitor loss contributes near the peak as well, where the shunt branch carries its largest current and any series resistance in the capacitor dissipates power that never reaches the load. The 680 and 2200 Ω sweeps have broader peaks or fall at cleaner nominal values, so they are less sensitive to these effects and track theory more closely.

One last observation is that the compensated network showed real voltage magnification. At $R_L = 2.2\text{ k}\Omega$ near the optimum capacitance the load amplitude reached 3.49 V from a 2.0 V source, about $1.7\times$. This is the lightly damped series resonance that makes overcompensation risky in practice, since pushing capacitance past the optimum can drive load voltages above their ratings even while delivered power falls.